

Quantum Mechanics II Midterm Exam 2

Instructions:

- Pick 3 out of the 4 questions below.
- Use natural units $\hbar = c = 1$.

Question 1) Consider the Yukawa potential,

$$V(\mathbf{r}) = \alpha_Y \frac{e^{-\mu r}}{r}. \quad (1)$$

- a) Calculate the Fourier transform $\tilde{V}(\mathbf{q})$ of the Yukawa potential, where $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ is the momentum transfer.
 (Hint: You will need the integral $\int_0^\infty dr e^{-mr} \sin(qr) = \frac{q}{m^2 + q^2}$, with $q = |\mathbf{q}|$.)
- b) Show that the differential cross section $d\sigma/d\Omega$ for the Yukawa potential using the Born approximation for the scattering amplitude

$$f(\theta) = -\frac{m}{2\pi} \tilde{V}(\mathbf{q}). \quad (2)$$

is given by

$$\frac{d\sigma}{d\Omega} = \frac{4\alpha_Y^2 m^2}{(4m^2 v^2 \sin^2(\theta/2) + \mu^2)^2}, \quad (3)$$

where v is the relative velocity of the incoming particle in the center-of-mass frame.

- c) Write your answer in the two limits: $\mu \ll q$ and $\mu \gg q$. Interpret the physical meaning of these limits in terms of the range of the potential. (Note: Your result for the differential cross section in the limit $\mu \gg q$ should be equivalent to using the contact interaction

$$V(\mathbf{r}) = \frac{4\pi\alpha_Y}{\mu^2} \delta^3(\mathbf{r}), \quad (4)$$

in the Born approximation.)

- d) What partial wave(s) contribute to the scattering amplitude in the limit $\mu \gg q$? Why?
- e) Take $\mu = m_W \simeq 100$ GeV and $\alpha_Y = \alpha_W \simeq 0.1$. Estimate the **total** cross section σ in the contact interaction limit for $m = 1$ GeV. Take $4\pi \simeq 10$. Give your answer in cm^2 (use $1 \text{ eV} \simeq 10^4 \text{ cm}^{-1}$).

Question 2) The ansatz solution for the incoming and outgoing scattering states in the potential scattering problem,

$$\psi(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} + f(\theta, \phi) \frac{e^{ikr}}{r}. \quad (5)$$

- a) Explain the physical meaning of each term in the ansatz solution, including the sign of both exponents. Give an example of a potential for which this ansatz solution is not strictly valid.
- b) The scattering amplitude $f(\theta)$ can be expanded as

$$f(\theta) = \sum_{\ell=0}^{\infty} (2\ell + 1) e^{i\delta_\ell} \sin \delta_\ell P_\ell(\cos \theta). \quad (6)$$

Starting from this formula, show that the optical theorem holds:

$$\sigma_{\text{tot}} = \frac{4\pi}{k} \text{Im} f(0). \quad (7)$$

(Hint: use the orthogonality property $\int_{-1}^1 d\mu P_\ell(\mu) P_{\ell'}(\mu) = \frac{2\delta_{\ell\ell'}}{2\ell+1}$ and $P_\ell(1) = 1$ for all ℓ .)

- c) Rewrite the Lippmann-Schwinger equation for the scattering state $|\psi_{\mathbf{k}}^{(+)}\rangle$,

$$|\psi_{\mathbf{k}}^{(+)}\rangle = |\phi\rangle + G_0(E)V|\psi_{\mathbf{k}}^{(+)}\rangle,$$

in position space in terms of the scattering solution $\psi_{\mathbf{k}}^{(+)}(\mathbf{x})$, the free plane wave solution $\phi(\mathbf{x}) = e^{i\mathbf{k}\cdot\mathbf{r}}$, the free particle Green's function $G_0(\mathbf{x}, \mathbf{x}'; E)$, and the potential $V|\mathbf{x}\rangle = V(\mathbf{x})|\mathbf{x}\rangle$.

- d) Derive the Born approximation for the scattering amplitude $f(\theta)$. Recall that the free particle Green's function in the energy domain in position space is given by

$$G_0(\mathbf{x}, \mathbf{x}'; E) = -\frac{2m}{4\pi} \frac{e^{ik|\mathbf{x}-\mathbf{x}'|}}{|\mathbf{x}-\mathbf{x}'|}.$$

(Hint: For large distances, $|\mathbf{x}-\mathbf{x}'| \simeq r - \hat{\mathbf{r}} \cdot \mathbf{x}'$. Define $\mathbf{q} = \mathbf{k}' - \mathbf{k}$ with $\mathbf{k}' = k\hat{\mathbf{r}}$.)

Question 3) Take the Rabi oscillation of a two-level system with a circular time-dependent drive

$$H(t) = \frac{\omega_0}{2} \sigma_z + \frac{\Omega}{2} \cos(\omega t) \sigma_x + \frac{\Omega}{2} \sin(\omega t) \sigma_y = \frac{1}{2} \begin{pmatrix} \omega_0 & e^{i\omega t} \\ e^{-i\omega t} & -\omega_0 \end{pmatrix}. \quad (8)$$

We define the detuning parameter $\delta = \omega - \omega_0$ and the generalized Rabi frequency $\Omega_R = \sqrt{\Omega^2 + \delta^2}$. Take the two eigenstates of σ_z as the basis states:

$$|+\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |-\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (9)$$

Also recall,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10)$$

- a) In the limit $\delta = 0$, using the interaction-picture potential,

$$V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t} = \frac{\Omega}{2} \begin{pmatrix} 0 & e^{i\delta t} \\ e^{-i\delta t} & 0 \end{pmatrix}, \quad (11)$$

calculate the probability for the system to be in the state $|-\rangle$ at time t if it starts in the state $|+\rangle$ at $t = 0$ using the exact time evolution operator in the interaction picture,

$$U_I(t) = \mathcal{T} e^{-i \int_0^t dt' V_I(t')} = e^{-i V_I t}. \quad (12)$$

(Hint: You will need the relation $e^{i\alpha \hat{n} \cdot \sigma} = \mathbb{1} \cos \alpha + i(\hat{n} \cdot \sigma) \sin \alpha$ for any unit vector $\hat{n}$. Recall that in the interaction picture, $|\psi_I(t)\rangle = e^{-iV_I t} |\psi(0)\rangle$.)

- b) What happens at $t = 2/\Omega$? Explain qualitatively how the result in (a) changes when $\delta \neq 0$.
- c) Imagine that instead of a circular drive, we had a linear and diagonal perturbation given by

$$H(t) = \frac{\omega_0}{2} \sigma_z + \frac{\Omega(t)}{2} \sigma_z. \quad (13)$$

State (with or without calculating) the probability for the system to be in the state $|-\rangle$ at time t to be if it starts in the state $|+\rangle$ at $t = 0$?

Question 4.

- a) Starting from the symmetric and antisymmetric two-particle states for two identical bosons or fermions,

$$|\psi_{\pm}\rangle = \frac{1}{\sqrt{2}} \left(|\phi_1\rangle_1 \otimes |\phi_2\rangle_2 \pm |\phi_2\rangle_1 \otimes |\phi_1\rangle_2 \right), \quad (14)$$

show that the probability density for finding one particle at position x_1 and the other particle at position x_2 is given by

$$|\psi_{\pm}(x_1, x_2)|^2 = \frac{1}{2} \left(|\phi_1(x_1)|^2 |\phi_2(x_2)|^2 + |\phi_2(x_1)|^2 |\phi_1(x_2)|^2 \pm 2\text{Re}[\phi_1^*(x_1)\phi_2^*(x_2)\phi_2(x_1)\phi_1(x_2)] \right). \quad (15)$$

- b) Quantitatively or qualitatively, how does this differ from the probability density for distinguishable particles?
- c) Consider the case where ϕ_1 and ϕ_2 are two Gaussian wave packets centered at positions a and b , respectively, with width σ ,

$$\phi_1(x) \sim e^{-\frac{(x-a)^2}{2\sigma^2}}, \quad \phi_2(x) \sim e^{-\frac{(x-b)^2}{2\sigma^2}}. \quad (16)$$

Calculate $|\psi_{\pm}(x_1, x_2)|^2$ for this case (do not worry about normalization factors and note that the wavefunctions are real).

(Hint: you may find it useful to define $c = \frac{a+b}{2}$ and $d = \frac{a-b}{2}$ and use

$$(x-a)^2 + (x-b)^2 = 2(x-c)^2 + 2d^2. \quad (17)$$

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- d) What happens in the limits $|d| \gg \sigma$ and $|d| \ll \sigma$ for fermions? And for bosons?